

- 1 A sequence $u_n, n \in \mathbb{Z}^+$ is defined by $u_{n+1} = \frac{n+2}{n^2} u_n$ and $u_1 = 2a$, where a is a fixed real constant. Prove by mathematical induction that

$$u_n = \frac{a(n+1)n}{(n-1)!} \text{ for } n \in \mathbb{Z}^+.$$

State the limit of the sequence.

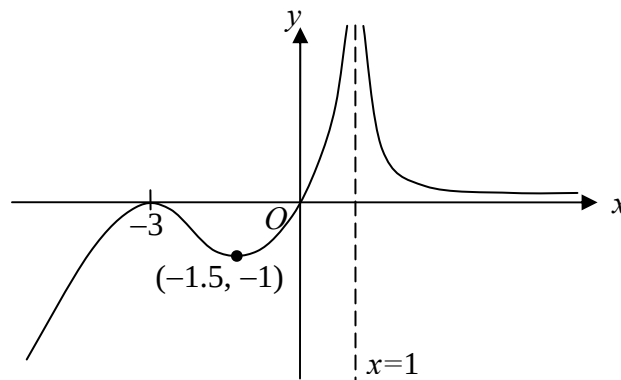
[5]

- 2 A curve C is defined parametrically by

$$x = 2 \sin t - \sin 2t, \quad y = 2 \cos t - \cos 2t, \quad \text{where } -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}.$$

- (i) Sketch the graph of C . [2]
- (ii) A flower bed is bounded by the curve C and the line $y = 1$. Find the area of the flower bed, giving your answer correct to one decimal place. [3]

- 3 The diagram below shows the graph of $y = f'(x)$. The curve passes through the origin and has turning points at $(-3, 0)$ and $(-1.5, -1)$. The x -axis and $x = 1$ are the two asymptotes of the curve.



- (i) Find the range of values of x for which the graph of $y = f(x)$ is strictly increasing and concave downwards. [1]
- (ii) State the x -coordinates of all the stationary points of the graph of $y = f(x)$ and determine the nature of each point. [2]
- (iii) Given that $f(0) = 1$, sketch the graph of $y = f(x)$ for $x < 1$. Your sketch should indicate clearly all stationary points, asymptotes and intersections with the axes. [2]

[Turn over]

- 4 A fixed complex number a is such that $0 < \arg(a) < \frac{\pi}{2}$. On a single Argand diagram, sketch the loci given by $|z - a| = |z - 7a|$ and $|z - 4a| = 3|a|$. [3]

The two complex numbers that satisfy the above equations are represented by the complex numbers p and q .

Find the possible values of $\arg\left(\frac{p}{q}\right)$. [3]

Find $|p + q|$ in terms of a . [1]

- 5 Express $g(r) = \frac{1}{(3r-2)(3r+1)(3r+4)}$ in the form $af(r) - bf(r+1)$ where a and b are real constants to be determined, and $f(r) = \frac{1}{(3r-2)(3r+1)}$. [2]

Hence or otherwise, find in terms of n ,

(i) $\sum_{r=1}^n g(r)$, [2]

(ii) the sum of the infinite series $\frac{1}{(3n-5)(3n-2)(3n+1)} + \frac{1}{(3n-2)(3n+1)(3n+4)} + \dots$. [2]

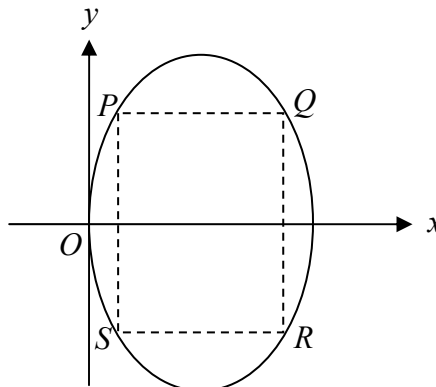
By considering the graph of $y = g(x)$, explain why $\sum_{r=1}^n g(r) > \int_1^{n+1} g(x) dx$. [2]

- 6 The curve C has equation $(ay)^2 = x(4-x)$, where $a > 0$.

(i) By differentiation, find the equations of the tangents which are parallel to the y -axis. [3]

(ii) The diagram of a land plot bounded by the curve C where $a = 0.5$ is shown below. A rectangular Jacuzzi pool, $PQRS$ is to be constructed where points P , Q , R and S are on C . Denoting P by (x, y) , show that area of $PQRS = 8(2-x)\sqrt{4x-x^2}$.

Hence find the exact value of x for the area to be maximum. [5]



[Turn over

- 7 Express $f(x) = \frac{3x^2 - 3x + 4}{(2x-1)(x^2+3)}$ in partial fractions and hence obtain $f(x)$ as a series

expansion in ascending powers of x up to and including the term in x^2 . [5]

(i) Find the range of values of x for which the expansion is valid. [1]

(ii) Find the coefficient of x^{2n} in the expansion, where n is a positive integer. [2]

- 8 The functions f and g are defined as follows:

$$f : x \rightarrow \frac{5-x}{1-x}, \quad x \in \mathbb{R}, x \neq 1,$$

$$g : x \rightarrow 2x^2 + 4x + \lambda, \quad x \in \mathbb{R}, x > -2.$$

(i) Explain why f has an inverse, f^{-1} , and show that $f^{-1} = f$. [4]

(ii) Evaluate $f^{51}(4)$. [2]

(iii) Find the range of values of λ such that fg exists.
For these values of λ , find the range of fg in terms of λ . [3]

- 9 (a) An arithmetic progression has first term a and common difference -1 . The sum of the first n terms is equal to the sum of the next $2n$ terms, where n is a positive integer. Find a in terms of n . [3]

- (b) Mr. and Mrs. Lee decide to invest some money to pay for their daughter's tertiary education.

Mrs. Lee makes annual contributions of \$1000 into an investment fund. She makes the first contribution on the day her daughter is born, and the last contribution on her daughter's seventeenth birthday. Her investment earns 5% per annum.

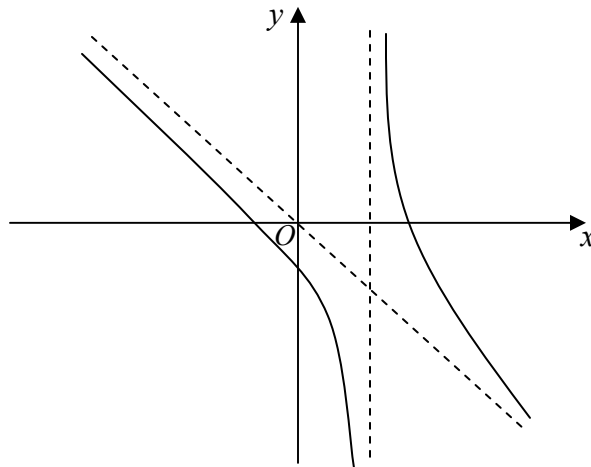
Mr. Lee makes contributions into another investment fund. He contributes \$500 on the day his daughter is born, and increases his annual contribution by 5% per annum until his last contribution on his daughter's seventeenth birthday. His investment also earns 5% per annum.

Interest for both Mr. and Mrs. Lee's funds are credited into the account annually on the next birthday of their daughter.

Find the amount in the investment fund of each parent on their daughter's eighteenth birthday. [7]

[Turn over

- 10 (i) Differentiate $\sqrt{1-x^2}$ with respect to x . [1]
- (ii) Find $\int \cos^{-1} x \, dx$ and hence evaluate $\int_{-1}^1 \cos^{-1} x \, dx$, giving your answer in exact form. [3]
- (iii) The finite region R is bounded by the curve
 $y = \cos^{-1}(x-2)$, where $1 \leq x \leq 3$,
the line $y = \pi$, the x -axis and y -axis.
- (a) Describe the geometrical transformation which maps the graph of $y = \cos^{-1} x$ onto the graph of $y = \cos^{-1}(x-2)$.
Hence find the area of the region R in exact form. [3]
- (b) Find the exact value of the volume generated when R is rotated through 2π radians about the y -axis. [4]
- 11 A sketch of the curve $y = f(x)$ is shown in the diagram below (not drawn to scale). The lines $x=1$ and $y=-x$ are asymptotes to the curve.



Given $f(x) = \frac{ax^2 + bx + c}{x + d}$, where a, b, c and d are constants, write down the value of d , and determine the values of a and b . [3]

- (i) Find the coordinates of the points (in terms of c) where the curve crosses the axes. Hence find the range of values of c . [3]
- (ii) Sketch the graph of $y = \frac{1}{f(x)}$, making clear the main relevant features of the curve.

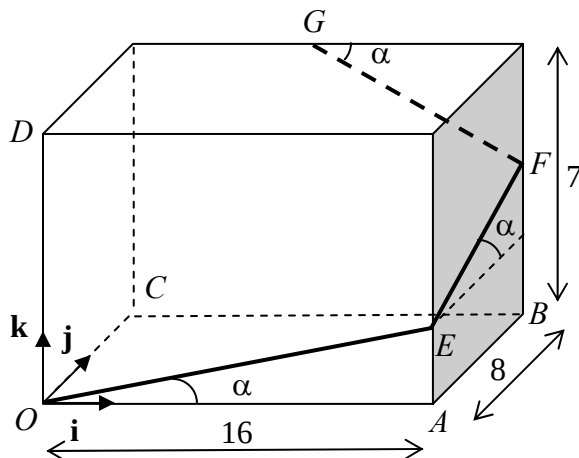
By sketching the graph of $y = \frac{1}{1-x}$ on the same diagram, solve the inequality

$$\frac{1}{f(x)} < \frac{1}{1-x},$$

leaving your answer in terms of c . [6]

[Turn over]

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In the above diagram, a rectangular box has base $OABC$. The point O is taken as the origin, with unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ in the direction of $\overrightarrow{OA}$, $\overrightarrow{OC}$ and $\overrightarrow{OD}$ respectively. The length of OA , OC and OD are 16 cm, 8 cm and 7 cm respectively. A spider crawls along the surfaces of the rectangular box. The route $OEF G$ taken by the spider starts at one corner, O , of the base of the box and continues over three surfaces of the box to end at a point G along an edge of the box. The line of the route always makes a fixed angle α with the horizontal, with $\tan \alpha = \frac{1}{4}$.

- (i) Find the position vectors of E , F and G . [2]
- (ii) Find the area of the triangle OEG . Hence find the distance from G to the line OE . [4]
- (iii) Find an equation of the plane that contains the points E , F and G in the form $\mathbf{r} \cdot \mathbf{n} = p$, where $\mathbf{n}$ and p are to be determined. [2]
- (iv) Determine whether the points O , E , F and G lie on the same plane. Justify your answer. [2]
- (v) A plane Π contains the points O and F . Given that the three planes Π , OEG and EFG do not have any point in common, find an equation of Π in the form $\mathbf{r} = \mathbf{u} + \lambda \mathbf{v} + \mu \mathbf{w}$, where $\lambda, \mu \in \mathbb{R}$. [2]

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